PING-AI state comparison

nb050 · 9 June 2026

Abstract

One network (COBANet) can run as the gamma rhythm **PING** or as the Vreeswijk-Sompolinsky **asynchronous-irregular** state. With everything else harmonised — connectivity, weights, W^{II} — the two differ in a single control: the **correlation of the input** across cells. Shared (correlated) drive gives gamma; per-cell independent drive gives the balanced state.

Methods

Model parameters

The two models are identical COBANet except for the input. The table is the full configuration; the only rows that differ — all of them describing the input — are in *italics*.

Parameter	PING	V&S AI
Model	COBANet (conductance LIF)	COBANet (conductance LIF)
N_E / N_I	1024 / 256	1024 / 256
Δt / T	0.25 ms / 1000 ms	0.25 ms / 1000 ms
τ_E / τ_I (AMPA / GABA)	2 / 9 ms	2 / 9 ms
Recurrent connectivity	sparse fixed-fan-in, $p = 0.2$	sparse fixed-fan-in, $p = 0.2$
W^{EI} (E→I)	$\mathcal{N}(0.6, 0.18)$ µS	$\mathcal{N}(0.6, 0.18)$ µS
W^{IE} (I→E)	$\mathcal{N}(3.0, 0.9)$ µS	$\mathcal{N}(3.0, 0.9)$ µS
W^{II} (I→I)	none	none
External drive	*dense W_{in} (shared 1024-neuron layer)*	*diagonal W_{in} (one private input per cell)*
Drive rate	*45 Hz (E)*	*45 Hz (E), 8 Hz (I)*
Input correlation across cells	*correlated*	*independent*

Both models use sparse fixed-fan-in recurrent connectivity at $p = 1 - s = 0.2$ on the E↔I matrices ($W^{EE} = 0$, W^{II} off). Because p is also the fraction of recurrent input any two cells share ($\langle n_{\text{shared}} \rangle / K = K/N = p$), it sets how strongly the loop can cohere — connectivity is the *internal* counterpart of the input knob. We fix it at a cortically realistic $p = 0.2$ (local E↔I connectivity runs ≈ 0.1–0.5) — dense enough for the recurrent loop to cohere, so connectivity is shared and does not distinguish the regimes.

Model differences

The models share everything above — connectivity, the E↔I weights, W^{II}. The single difference is the **input**: a shared (correlated) drive for PING, per-cell independent (uncorrelated) drive for the V&S cell.

Both deliver 45 Hz Poisson through the *same* input pathway — spikes routed by W_{in} into a decaying AMPA conductance. The only difference is the **structure of W_{in}**. The external excitatory conductance onto E cell j obeys

$$\tau_E \dot{g}_j^{\text{ext}}(t) = -g_j^{\text{ext}}(t) + \sum_k W_{jk}^{\text{in}} s_k(t), \quad W^{\text{in}} = \begin{cases} \text{dense} & \text{(PING)} \\ g_0 \mathbb{1} \text{ (diagonal)} & \text{(V\&S AI)} \end{cases}$$

where:

- $g_j^{\text{ext}}(t)$ — external excitatory conductance onto E cell j;
- τ_E — AMPA decay time constant (2 ms), common to both;
- $s_k(t) = \sum_m \delta(t - t_k^m)$ — spike train of input neuron k (Poisson, 45 Hz); $N_{\text{in}} = 1024$;
- W^{in} — input weight matrix: **dense** for PING (every E cell reads the whole input layer, so all cells share the same drive) or **diagonal** $g_0\mathbb{1}$ for the V&S cell (each E cell reads only its own input neuron, $W_{jk}^{\text{in}} = g_0\delta_{jk}$, so the drives are independent);
- g_0 — per-spike conductance, 0.38 µS (the diagonal weight).

A dense W_{in} correlates the input across cells; a diagonal W_{in} leaves it independent. (In the runner the diagonal case is realized as a per-cell independent conductance rather than a literal identity matrix — the two are equivalent.)

The consequence is the cross-cell correlation of the drive:

$$\mathrm{Corr}\Big(g_j^{\text{ext}}, g_{j'}^{\text{ext}}\Big) > 0 \quad \text{(PING, shared layer)}, \quad \mathrm{Corr}\Big(g_j^{\text{ext}}, g_{j'}^{\text{ext}}\Big) = 0 \quad \text{(V\&S AI, private streams)}, \quad j \neq j'.$$

The shared layer is a *common reference* the loop can lock onto; the private streams kick each cell independently, with no common phase. Both drive E at the same 45 Hz, so rate is not the difference. What both share: neither input carries any rhythm. Both are featureless, steady Poisson (Figure 1), so PING's gamma is generated by the network, not inherited from the drive.

Results

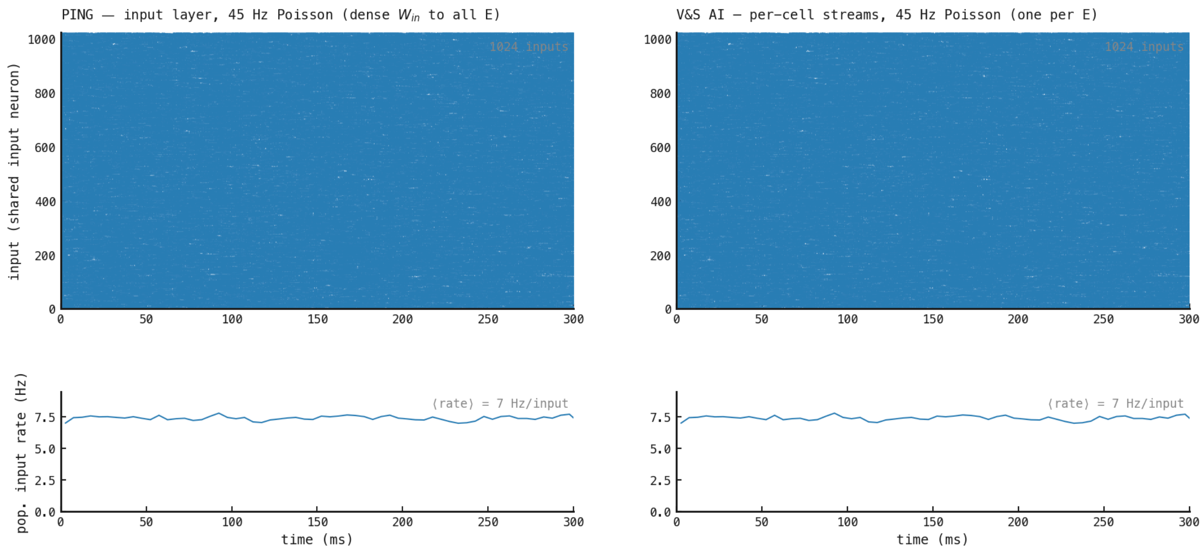

Figure 155: Top: a 300 ms raster of the drive — PING's shared input layer (left) and the V&S cell's per-cell private streams (right), both 45 Hz. Bottom: the population rate of each drive. Both are featureless, steady Poisson at the same rate; the input carries no oscillation in either regime. The regimes differ only in *correlation structure* — shared layer vs private streams — not in rate or temporal content.

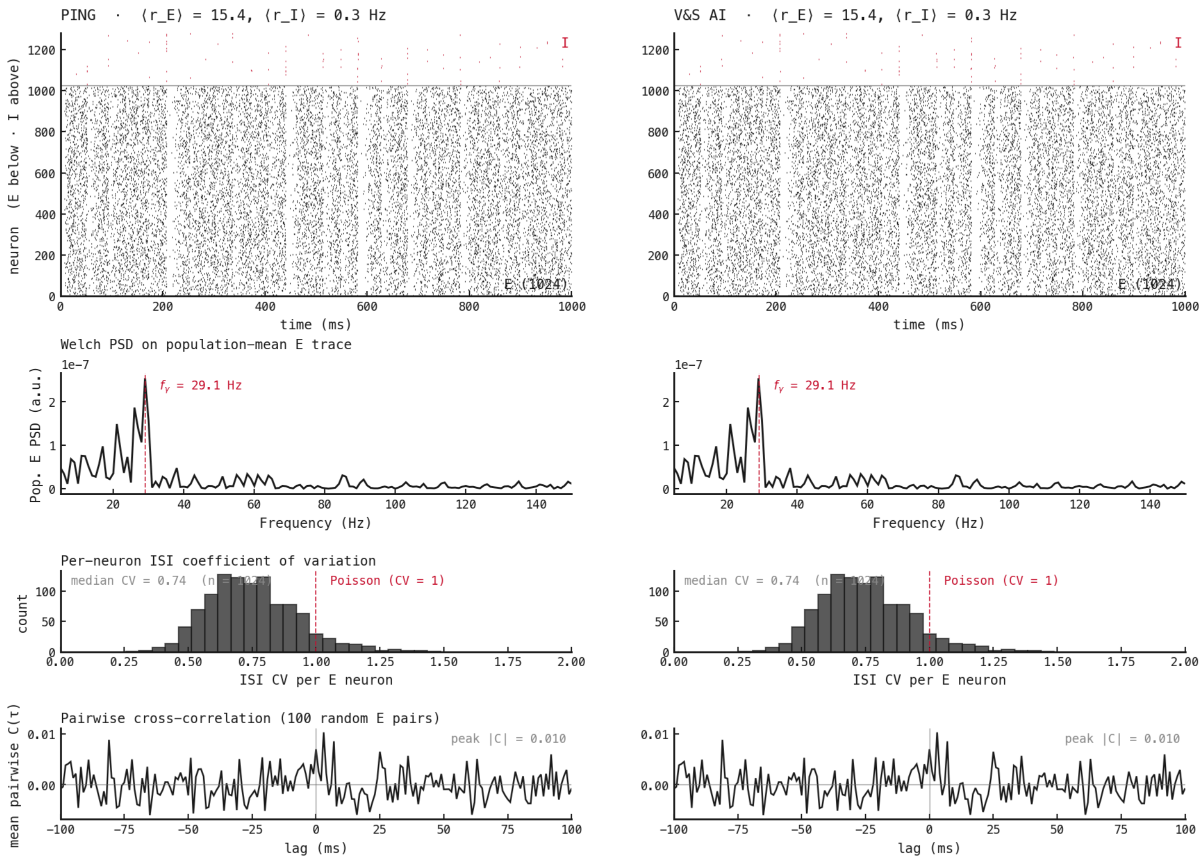

Figure 156: Row 1 is the raster with both populations on one axis — E (black) below, I (red) above the divider — so the I-leads-E gamma timing is visible directly. Both networks have identical wiring (sparse $p = 0.2$, matched weights, no W^{II}); only the input differs. Left, PING (shared input): cycle-locked gamma bursts, a sharp 32 Hz peak, regular (sub-Poisson) ISIs (CV 0.64), cross-correlations combing at the gamma period (peak 0.063). Right, V&S AI (independent input): scattered raster, broadband spectrum, irregular ISIs, flat cross-correlation (peak 0.010). Same network and wiring, opposite dynamics — the input correlation alone.

Next steps

The two regimes are now identical except for the input — same connectivity, weights, and 45 Hz E-drive rate — and the difference is its **cross-cell correlation**: a shared input layer (PING) versus per-cell independent streams (V&S). (One residual asymmetry: the V&S cell drives its I pool with an independent 8 Hz stream to decorrelate it, where PING's I is loop-driven.) Turning correlation into a single, continuous control is the remaining work.

A faithful correlation knob. The tidy idea — mix a shared common stream with private streams and dial the shared fraction $c \in [0, 1]$ — breaks at the $c = 1$ extreme if "shared" is *one* stream broadcast verbatim: every cell then receives identical input and locks into trivial synchrony (cross-correlation ≈ 0.99, a broad ≈ 74 Hz peak) rather than loop-generated gamma. Real correlated drive is a shared *population*, not one identical signal. The fix is a shared *layer* of K_{shared} common input neurons (each cell reading all of them), interpolated against private streams at matched rate — correlated without being identical, so genuine gamma survives. With that knob, the gamma↔asynchronous boundary can be mapped continuously over c.